

Sandeep Poloju

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SUMMARY

Software Engineer with 3+ years shipping production distributed systems across fintech, govtech, and research domains. Builds full-stack applications with Angular/React, Spring Boot, and FastAPI, and integrates intelligent features using LLMs, RAG pipelines, and agentic AI. Proven track record of owning problems end-to-end across system design and DevOps (AWS, Kubernetes) in fast-paced Agile teams.

EXPERIENCE

GovieRates

Mar 2026 – Present

Software Developer Intern

Largo, MD

- Built a middleware REST API layer using **Java Spring Boot** that acts as a data sync bridge between the GovieRates accounting platform and Project Magnus (a DCAA-compliant timesheet system), eliminating data mismatches across client accounts.
- Built an end-to-end client onboarding flow integrating SSO JWT entry from GovieRates, Stripe subscription payment with a **7-tier pricing model**, and automated multi-tenant provisioning, onboarding **30+ clients** and growing subscriptions from **\$1,000 to \$5,000/month**.
- Designing an agentic AI layer using **LangGraph** to enable natural language querying and intelligent automation across the GovieRates and Project Magnus platforms.
- Tech Stack: **Next.js 15, Java Spring Boot, MongoDB Atlas, JWT, Stripe, LangGraph, Azure Container Apps, SendGrid**

University of Maryland

Apr 2024 – Jan 2025

Research Assistant (Software Engineer)

College Park, MD

- Developed a full-stack AgriTech web application to predict Brown Patch disease severity, assembling a **10GB+ training dataset** by unifying data from PDFs, websites, and NOAA weather data from **AWS S3**.
- Improved model accuracy **from 74% to 91%** through Köppen climate-based regional segmentation and stepwise feature selection; trained Random Forest and Neural Network models using **AWS SageMaker**.
- Deployed serverless inference via **AWS Lambda and API Gateway** for cost-efficient scaling.
- Tech Stack: **Python, TesseractOCR, BeautifulSoup, REST APIs, AWS (S3, Lambda, API Gateway), TensorFlow, Scikit-learn**

Infosys

May 2021 – Dec 2023

Software Engineer

Hyderabad, India

- Diagnosed a latency bottleneck in the search suggestion API during development, profiled the Cypher query and serialization layer, re-architected both with optimized queries and Protobuf, reducing response time from **~200ms to ~30ms per keystroke**.
- Integrated LLMs to boost NLU search accuracy from **85% to 92%** and designed a Graph Neural Network (GNN) microservice for link prediction on a financial services platform.
- Simplified microservice data serialization to create a minimal API payload, cutting network traffic by **68% (465KB to 148KB)**.
- Engineered a Python translation layer within a core microservice to convert GraphQL queries directly to Cypher, bypassing an inefficient intermediate service and improving system performance.
- Engineered RESTful APIs for User Preferences using **Node.js and PostgreSQL**, implementing RBAC to serve role-specific configurations across admin and employee dashboards.
- Designed a dynamic multi-tenant routing layer using **NGINX** reverse proxy, eliminating an intermediate routing microservice and reducing infrastructure complexity.
- Refactored a legacy Angular front-end using **SOLID principles** and the Adapter Design Pattern; standardized WCAG 2.1 Level AA accessibility, resolving **40+ UI barriers**.
- **Led a team of three** to automate the regression testing pipeline, eliminating **20 hours** of manual effort per release.
- Tech Stack: **Angular, TypeScript, Python, Java, Node.js, Spring Boot, RESTful APIs, PostgreSQL, Neo4j, Cypher, GraphQL, LLMs, NGINX, Docker, Kubernetes**

PROJECTS

- Student Q&A RAG System** | *FastAPI, LangGraph, MCP, ChromaDB, Docker, Ollama* | [GitHub](#) Feb 2026
- Built a RAG pipeline over **50,000+** academic PDFs using **LangGraph** for stateful orchestration, which classifies, retrieves, and rejects off-topic queries via conditional routing.
 - Designed a provider-agnostic LLM layer switchable between cloud (Nebius) and local Ollama inference for air gap deployment; containerized with Docker for one-command portable deployment.
 - Integrated a **Model Context Protocol (MCP)** server for dynamic web search via DuckDuckGo, enabling LangGraph to route off-topic queries to live internet results instead of rejecting them.
- GREBoost** | *Angular, TypeScript, Flask, MongoDB, Vercel, GitHub Actions* | [GitHub](#) Sep – Nov 2024
- Developed a microservices-based e-learning platform, implementing RESTful APIs and an Angular frontend, deployed on Vercel with automated CI/CD using GitHub Actions.

EDUCATION

University of Maryland, College Park College Park, MD
Master of Engineering, Software Engineering — GPA: 3.93 Jan 2024 – Dec 2025

CERTIFICATIONS

- **Claude Code in Action** by Anthropic | [Credential](#)

ACHIEVEMENTS & PUBLICATIONS

- **Top 100 Rank (Top 0.06%)**, HackwithInfy National Coding Competition (167,000+ participants).
- **Runner-up among 95 teams** in the WatsonX Chatbot Challenge | [Submission](#)
- Competitive programmer: Highest rating CodeChef 1799, CodeForces 1370.
- **Solved 600+** advanced algorithmic problems across major platforms ([LeetCode](#), [CodeForces](#), [CodeChef](#), [HackerRank](#)).
- Published: "Analyzing Twitter Event Hype vs. Reality" – NLP sentiment analysis (IJRAR) | [Paper](#)
- Cleared **Amazon SDE interview loop**.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Cypher (Graph Query), Shell Scripting
AI/ML: RAG, LangChain, LangGraph, OpenAI/Gemini APIs, PyTorch, TensorFlow, NLP, MCP
Frontend: Angular (Expert), React, Next.js, Flutter, HTML5, SCSS, RxJS
Backend: Spring Boot, FastAPI, Flask, GraphQL, gRPC
DevOps: AWS (Lambda, SageMaker, EC2), Azure Container Apps, Docker, Kubernetes, NGINX, GitHub Actions, Vercel, CI/CD
Databases: PostgreSQL, MongoDB, Neo4j, Redis
Tools: Cursor, Claude Code, Monday.com, Git, Postman